

Surgery for upper urinary tract urothelial cancer (UTUC)

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About surgery for upper urinary tract urothelial cancer

Surgery is usually the main treatment for [UTUC](#) that has not spread to other parts of the body.

Surgery may be open or keyhole (laparoscopic).

In general, if you have keyhole surgery you may not need to stay in hospital for as long. Recovery from keyhole surgery is usually quicker than from open surgery.

Open surgery

During open surgery, the surgeon makes 1 or 2 large cuts (incisions).

Keyhole (laparoscopic) surgery

During keyhole surgery, the surgeon makes 3 to 5 small cuts. They pass a telescopic camera (laparoscope) through one of the cuts. It sends video images to a monitor. The surgeon will then pass special surgical tools through the other cuts to remove the cancer. One of the cuts may be bigger so the surgeon can remove the kidney and ureter.

Image: Image shows patient lying on a surgical table with a cover over the body. There is a square of material cut out to show the abdomen (tummy). The surgeon is using a telescopic camera called a laparoscope through one of the small cuts in the abdomen. The surgeon is watching this on a monitor.

The surgeon may also use a specialised machine to help with keyhole surgery. The machine has cameras and very small surgical tools attached to robotic arms which the surgeon controls. This is called robot-assisted surgery. It makes it possible for the surgeon to do very precise surgery.

Some types of keyhole surgery may only be available at larger cancer centres. This means you may have to travel for treatment. Your surgeon will talk to you about which type of surgery is best for you.

Removing the kidney and ureter (nephroureterectomy)

This is the most common operation to treat UTUC. The surgeon removes:

- [the kidney](#)
- a layer of fat around the kidney
- [the ureter](#)
- an area of tissue where the ureter enters the bladder (bladder cuff).

The surgeon may also remove some [lymph nodes](#) close to the kidney to check whether they contain cancer cells.

You will have a tube (catheter) put in during the operation to drain urine from your bladder. This may be removed before you leave hospital. But you may go home with it to let your bladder heal.

You may have 1 dose of chemotherapy into the bladder after your surgery. This is to help reduce the risk of cancer coming back. This is called [intravesical chemotherapy](#). If you have a low risk of the cancer coming back, you usually do not need chemotherapy.

Removing a kidney and ureter is a big operation. It is not suitable for everyone. Your doctors may need to do tests to check you are well enough to cope with the surgery.

Removing part of the kidney

If you have 1 small tumour and there are no signs the cancer has spread, the surgeon may be able to remove part of the kidney only. This is called kidney-sparing surgery. It may also be an option for people who only have 1 kidney or if the kidneys do not work well.

Removing part of the ureter

Sometimes, if the cancer is in the lower third of the ureter, it may be possible to only remove part of the ureter. This is called a segmental ureteric resection.

The surgeon removes:

- the part of the ureter where the tumour is
- a healthy area (margin) of ureter above the tumour
- all the ureter below the tumour.

Then they reattach the remaining ureter to the bladder.

Endoscopic surgery

It is sometimes possible to remove a low grade, early stage tumour in the renal pelvis using endoscopy. The surgeon passes a flexible tube with a camera called an endoscope up through the urethra, bladder and ureter to the renal pelvis. This is called a ureteroscopy. The surgeon then removes the tumour using tools passed through the endoscope.

Sometimes the surgeon makes a small cut in your back to reach the kidney. This is called a percutaneous renoscopy. They then pass the endoscope into the kidney to reach the renal pelvis.

If you only have 1 kidney, you may have endoscopic surgery for higher grade cancers. Or you may have it if your kidneys do not work well enough for you to cope with just 1 kidney.

After surgery

Some people will have cancer drugs after surgery. This is called adjuvant treatment. The aim is to reduce the risk of cancer coming back. We have more information about [cancer drugs for UTUC](#).

If you have had part or all of a kidney removed, your remaining kidney tissue and other kidney will usually be able to do the work of both kidneys. Your cancer team will check this is happening with regular blood tests.

Possible complications of surgery

The most common complications after surgery are:

- a wound infection
- bleeding
- a chest infection
- a blood clot.

The nurses will monitor you. They will give you injections to reduce the risk of blood clots developing. Tell them straight away if you:

- have fevers or feel unwell
 - have any bleeding
 - notice swelling and redness in a limb (arm or leg)
 - have any symptoms of an infection, such as a cough or a leaking wound.
- You will be given a telephone number to call if you need advice when you get home.

Going home after surgery

How long you stay in hospital depends on the type of operation you have and how quickly you recover.

After keyhole surgery, most people go home after 2 to 5 days.

After open surgery, most people go home after 5 to 7 days.

Your body uses a lot of energy to heal. It is common to feel more tired than usual for about 6 weeks after the operation. Most people feel fully recovered after about 12 weeks.

You will not be able to do some everyday activities while you recover. For example, you may not be able to shop, drive, have sex, play sport or work. Your nurse or surgeon can tell you when you will be able to do these things again.

Before you leave hospital, you will be given an appointment for a check-up. This is usually about 6 weeks after your operation. It will be at an outpatient clinic. The appointment is a good time to ask questions or talk about any problems you have after your operation.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our UTUC information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Rouprêt M, Gontero P, Birtle A et al. European Association of Urology Guidelines on Upper Urinary Tract Urothelial Carcinoma. 2023. Available from <https://uroweb.org/guidelines/upper-urinary-tract-urothelial-cell-carcinoma>

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community



Chat online anonymously to others who understand what you are going through. Our

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

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